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Conceptual modelling to assess the influence of hydro-climatic variability on runoff processes in data scarce semi-arid Andean catchments

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ABSTRACT

Using hydro-meteorological time series of 50 years and *in situ* measurements, the dominant runoff processes in perennial Andean headwater catchments in Chile were determined using the hydrological model HBV light. First, cluster analysis was used to identify dry, wet and intermediate years. From these, sub-periods were identified with contrasting seasonal climatic influences on streamflow. By calibrating the model across different periods, impacts on model performance, parameter sensitivity and identifiability were investigated, providing insights into differences in hydrological processes. The modelling approach suggested that, independently of a dry or wet period of calibration, the streamflow response is mostly consistent with flux from groundwater storage, while only a small fraction comes from direct routing of snowmelt. The variation of model parameters, such as the groundwater rate coefficient, was found to be consistent with differing recharge in wet and dry years. The resulting snowmelt–groundwater model is a realistic hypothesis of the hydrological operation of such complex, data scarce and semi-arid Andean catchments. This model may also be a useful tool for predictions of seasonal water availability and a basis for further field studies.

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1 Introduction

In many parts of the world, headwater catchments at high elevations play a crucial role in supplying water to semi-arid regions downstream, where intensive irrigated agriculture is very common (Price and Egan 2014). Therefore, understanding the feedbacks of integrating climate, cryosphere and hydrology processes in such headwater regions is indispensable for short- and long-term water resources management. Climate change-induced temperature increases and changes in precipitation patterns are expected to have strong impacts on the regional hydrological cycle in many mountain environments, particularly in drought-prone areas, where water supply is predominantly generated by melting snow or ice (Bates *et al.* 2008). Barnett *et al.* (2005) highlight the particular vulnerability of the Andean, Himalayan and Hindu Kush headwater catchments to climate variability and change. In central Chile, export-oriented irrigated agriculture is the main economic activity (Álvarez *et al.* 2006, Oyarzún 2010). Since this region relies on glacier- and snow-derived water from the Andes, its population and economy are extremely vulnerable to climate change impacts and hydrological extremes (Barnett *et al.* 2005, Souvignat *et al.* 2010). We note that northern

central Chile endured a severe drought period between 2007 and 2014, with extremely low or below average precipitation rates.

Besides the notorious data scarcity that is generally evident in semi-arid regions of Chile, very few hydro-meteorological data are available for the Cordillera above elevations of around 1250 m. This imposes a major limitation on fulfilling the increasing demand for consistent long-term and short-term water availability predictions and scenarios (Souvignat *et al.* 2012). High-elevation data and information are particularly important to adequately represent snow-related processes in models in which climate, cryosphere and hydrological feedbacks are to be successfully simulated (Pellicciotti *et al.* 2008, Ragetti *et al.* 2014). Semi-arid regions are usually challenging environments for rainfall–runoff modelling; this is due partly to data scarcity and the significance of infrequent large precipitation events, but also to often unknown partitioning of contributions from near-surface, sub-surface and groundwater flow paths (Pilgrim *et al.* 1988, El-Hames and Richards 1994, Lidén and Harlin 2000, van Loon and van Lanen 2012). Therefore, relatively few hydrological modelling exercises have been carried out in semi-arid

mountainous catchments, e.g. Alpine (Bocchiola *et al.* 2011, Chauvin *et al.* 2011). Consequently there is little guidance on the appropriateness of modelling approaches to be applied for such environments, despite the crucial importance of this issue.

To date, only the Water Evaluation And Planning tool (WEAP) has been applied to headwater catchments in semi-arid northern central Chile north of 31.20°S latitude, where precipitation rates tend to be below 150 mm per year on average (Vicuña *et al.* 2010). WEAP is a widely used water system analysis and management model developed by the Stockholm Environment Institute, with a conceptual, semi-distributed rainfall–runoff hydrology module, which has been applied at the monthly scale by Vicuña *et al.* (2010). The representation of the catchment relies on a two-bucket system: the upper one generates streamflow dependent on rainfall and through the soil, while the lower bucket represents the baseflow processes through the deep soil or groundwater. Snow is calculated through a melting parameter, which is distributed across 100 m elevation bands via monthly temperature and precipitation (Ragettli *et al.* 2014). The model was used to analyse the impacts of future climate projections from statistical downscaling on streamflow changes, with the result that increasing temperatures and changing precipitation patterns are likely to significantly reduce long-term runoff due to accelerated snow and glacier melting and hence lead to lower water availability, especially in the summer time (Vicuña *et al.* 2010).

For wetter and glaciated catchments further south, WEAP has also been applied (Ragettli *et al.* 2014), as well as the physically-based, semi-distributed TOPKAPI-ETH and SWAT models based on more comprehensive datasets and observations (Pellicciotti *et al.* 2008, Stehr *et al.* 2010).

However, a simpler conceptual model at finer (e.g. daily) temporal resolution has not been applied to such a data scarce, complex environment and may represent a feasible alternative approach. Despite criticisms that conceptual models rely on calibration and that their parameters cannot be measured, they often use a more limited number of parameters compared to physically-based, distributed models and are less data intensive. Results can therefore often be used to provide insights into parameter identifiability and uncertainty (Uhlenbrook *et al.* 1999, Beven 2001a, 2001b). Also, empirical observations show that some physical aspects of high-elevation environments, such as limited vegetation cover and sparse soil formation, have a rather limited influence on the hydrological system, and hence they do not necessarily require complex model representation.

HBV is a commonly used conceptual model that has been developed and extensively applied in snow-dominated hydrological systems (e.g. Seibert and Vis 2012). Its application in semi-arid regions has been limited essentially to regionalization studies of ungauged mesoscale catchments (Love *et al.* 2010, Masih *et al.* 2010). Following recent work that used simple conceptual models and incorporated landscape classification and topography (Savenjie 2010, Fenicia *et al.* 2014) or tracer data (Soulsby *et al.* 2007, Hrachowitz *et al.* 2011), an open question is whether these approaches are appropriate for semi-arid, high-altitude catchments. Headwater catchments in central Chile provide an opportune test bed for simple, conceptual models as they are representative of such data scarce, snowmelt-driven high-elevation environments. Positive application of such models could indicate that they may be potentially transferable to other semi-arid areas and may be useful to assess irrigation and drinking water resources and, thus, food security in such regions (Barnett *et al.* 2005).

Here, we test HBV as a relatively simple, semi-distributed conceptual model in two Andean mesoscale headwater catchments (572 and 242 km²). The overall objective is to assess the effect of hydro-climatic variation on the hydrological function of the two catchments. Our specific objectives are: (a) applying the HBV-light rainfall–runoff model to these Andean catchments, characterized by extreme elevation and climatic gradient; (b) using statistical methods to separate dry and wet periods for model calibration; and (c) analysing the effects of climatic variability on streamflow uncertainty to support water and drought management.

2 Study area

The two gauged Andean headwater catchments used for modelling are the Rio Grande (572 km²) and the Tascadero (242 km²), which belong to the central Chilean Limarí River Basin located at latitude 30°15'–31°30'S and longitude 70°15'–71°45'W, covering an area of 11 696 km². The two sub-catchments were selected due to their long-term (>40 years) availability of discharge and climate data, in contrast to ungauged neighbouring sites. The elevation of the Rio Grande catchment ranges from 1381 to 4375 m, with a mean elevation of 3092 m, and that of the Tascadero from 1215 to 3847 m with a mean of 2889 m. The entire Limarí Basin reaches from the Pacific Ocean to the Andes up to an altitude of 5500 m (Fig. 1), and includes one of the most important regions in Chile in terms of production of high-value crops and the

extent of its irrigated agricultural surface area of 285 km² (Vicuña *et al.* 2014), and total potential agricultural area of 620 km² (Álvarez *et al.* 2006, Oyarzún 2010). The geology of the Limarí Basin consists of a variety of rock formations, ranging in age from Palaeozoic to Quaternary. Its cordillera belongs to the volcanic series of the Dona Ana Formation (Upper

Oligocene–Lower Miocene). Unlike other parts of the Andes, there is no Quaternary volcanism in the study region (Sernageomin 2012). Land cover is dominated by sparse shrub vegetation (Matorral con Suculentas) with cacti on bare ground, which can also be referred to as steppe, while above 2000 m the area is dominated by exposed rocks or weathered materials without

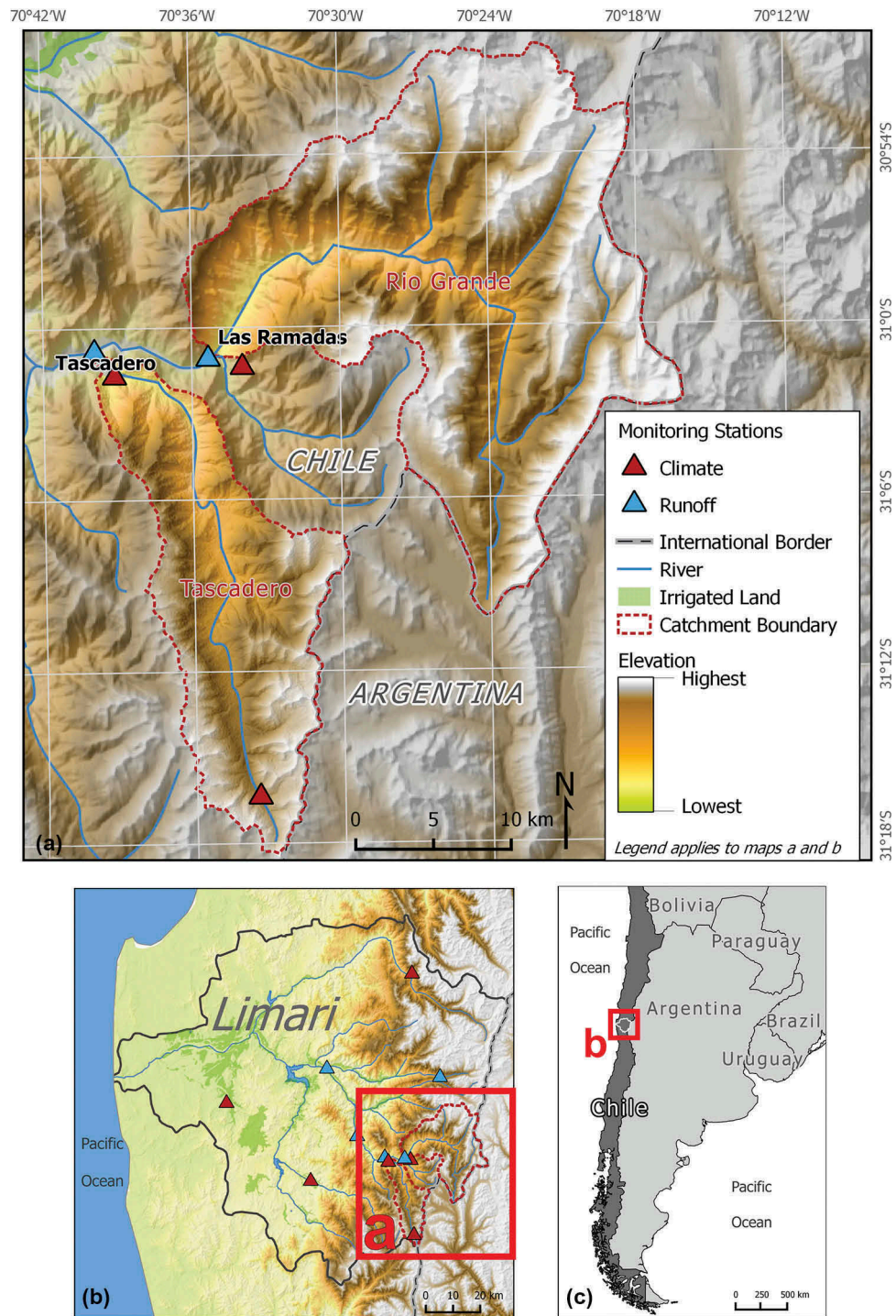


Figure 1. (a) The topography and main river courses of the Rio Grande (gauged at Las Ramadas station) and the Tascadero (Tascadero station) headwater catchments and monitoring stations within the (b) Limarí River Basin and (c) the regional context.

vegetation and some high-elevation peatlands in the alluvial riverbed areas (Schitteck *et al.* 2012, Nauditt *et al.* 2016). In the Andean area, shallow and unfertile eutric and dystic leptosols are the dominant soil cover (FAO 2002, DGA 2004).

The climate in the Limarí River Basin is classified as arid to semi-arid, as precipitation is much lower than potential evapotranspiration (Peel *et al.* 2007). However, the inter-annual variability of hydro-climatic conditions is high. Moreover, there is a significant correlation between the El Niño–Southern Oscillation (ENSO) signal and climate in central Chile (Montecinos and Aceituno 2003, Garreaud *et al.* 2009, Verbist *et al.* 2010, Meza 2013). La Niña years lead to dry conditions in central Chile during winter and late spring. On the other hand, higher El Niño sea surface temperatures (SSTs) lead to above-average rainfall in the eastern Pacific and central Chile during winter and late spring. Regional climate is driven by the Southeast Pacific anticyclone, the cold Humboldt Current along the Pacific coast and the steep topography of the mountain range of the Andes (Kalthoff *et al.* 2006, Oyarzún 2010). Westerly winds dominate above 4000 m, while below this elevation north–south winds tend to blow along the mountain range (Kalthoff *et al.* 2002). Each year, the study region faces a period of up to 10 months with no precipitation from September to June, with the mean annual precipitation ranging from 25 mm at the coast to 300 mm at Las Ramadas station at 1250 m (Favier *et al.* 2009, Souvignet *et al.* 2012).

Figure 2 shows the strong over-annual variability of monthly precipitation and discharge for the whole period of record (1968–2013). While the maximum measured monthly precipitation was 514.2 mm (June 1997), the mean monthly precipitation depth between

1968 and 2013 was only 25.3 mm. For the same period, the highest measured monthly discharge was 188.7 mm (November 1987), with a mean of 19.9 mm. Hydro-meteorological variations are common and groups of wet and dry years tend to cluster. The most significant droughts were observed in 1969–1972 and in 2008–2014, whereas the wettest years were 1987 and 1997, both associated with El Niño events. However, as snowfall occurs above 1200 m, a large proportion of the streamflow is generated during the spring and summer months (i.e. September to January) when snowmelt occurs.

Temperatures are strongly linked to radiation intensity, especially at high elevations (Kull *et al.* 2002, Favier *et al.* 2009). At Las Ramadas station, the mean annual temperature for 1968–2013 was 16.4°C with a maximum of 39.0°C and a minimum of −7.2°C. In the neighbouring Elqui River Basin (Cerro Tapado station), Kull *et al.* (2002) reported a mean annual temperature of −0.4°C, a maximum mean temperature (measured during the day) of 10°C and minimum mean temperature of −12°C at 4215 m elevation during hydrological year 1998/99. These values are consistent with recent measurements made by the authors in 2012 and 2013 at an elevation of 3250 m in the Tascadero catchment.

Unfortunately, no time series of snow-water equivalents are available for the region. The Limarí River Basin is non-glaciated, with estimated minor rock glacier coverage of 16.8 km², yet this is almost all outside the two study basins (Azócar and Brenning 2010). Favier *et al.* (2009) interpreted MODIS (MOD10A2) satellite data for the whole region and results showed that snow cover above 3000 m was present for four months on average. Seasonally measured discharge at high elevations, climate

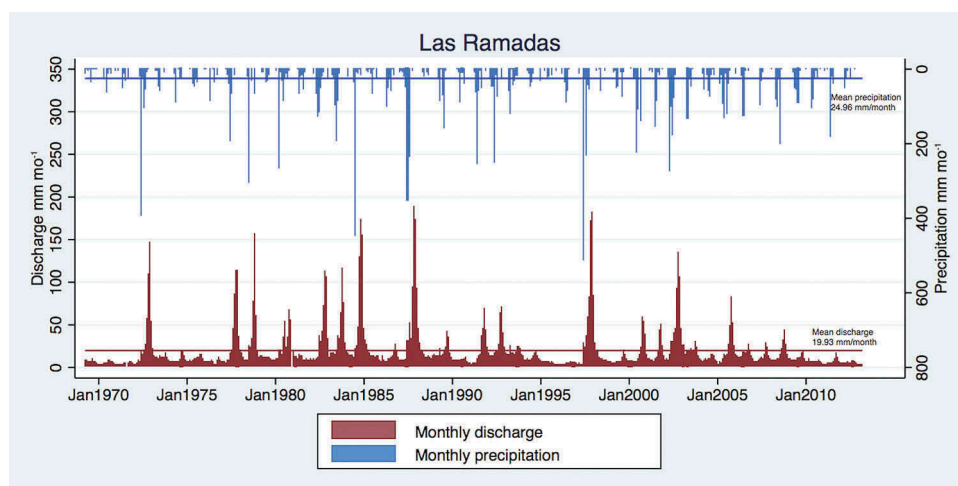


Figure 2. Time series of monthly precipitation and discharge data at Las Ramadas in the Rio Grande compared to the long-term average (1968–2013).

monitoring and sampling of geochemical and isotopic tracers in discharge indicate that runoff during the dry season originates exclusively from groundwater, which is recharged by snowmelt (Nauditt *et al.* 2016).

3 Data and methods

3.1 Data

Daily time series of 45 years (1968–2013) of precipitation, temperature and discharge for the Tascadero and Las Ramadas stations were collected by the Chilean water authority. Meteorological and discharge stations are located at the outlets of the Tascadero and Rio Grande headwater catchments. Discharge monitoring is carried out using automatic stage recorders in natural river sections, based on weekly calibration of $Q-h$ relationships by the DGA. Maintenance and data quality have improved during the past decade, while earlier records include periodic gaps and probably have larger rating curve errors during drier and wetter periods.

Meteorological time series were corrected by applying the inverse distance weighting method, while discharge gaps were filled using linear regression with data from neighbouring gauging stations. However, not all data errors in discharge time series could be captured by this method, as will be shown later. For altitudes above 1250 m, no long-term hydro-meteorological time series are available. Mean monthly potential evapotranspiration (ET_0) was provided by the Centro de Información de Recursos Naturales (CIREN) for the Las Ramadas station, at an elevation of 1250 m (Fig. 1). The ET_0 was calculated using the Penman-Monteith formula (CNR 1997). A 30 m digital elevation model (DEM) was obtained from the U.S. Geological Survey (USGS) and used to classify elevation zones using the open source Quantum Geographical Information System. Catchment characteristics are summarized in Table 1. The information on soils, vegetation, geology and upstream discharge used in the calibration procedure has been obtained from several field work campaigns since 2012 (Nauditt *et al.* 2016).

3.2 Statistical analysis

The identification of dry, medium and wet years within the time series was performed using a k -means cluster analysis (MacQueen 1967, Hartigan and Wong 1979) applied to daily discharge time series from the period 1968–2013. The clustering algorithm attempts to find groups (clusters) of similar elements from a dataset. Input for the k -means algorithm is the number of clusters desired, which in our case was three. The

Table 1. Basic characteristics (topography, climate and hydrology) for the Rio Grande and Tascadero headwater catchments.

Descriptor	Units	Rio Grande (Las Ramadas station)	Tascadero (Tascadero station)
Area	km ²	572	242
Elevation min	m.a.s.l.	1381	1215
Elevation max	m.a.s.l.	4375	3847
Mean elevation	m.a.s.l.	3092	2889
Mean slope	degrees	23.5	23.4
Mean river slope	degrees	9.6	7.0
Drainage density (Scale 1:200000)	km km ⁻²	0.085	0.178
Mean precipitation at climate station	mm	307	259
Mean annual temperature (catchment averages)	°C	5.1	6.29
Mean annual runoff	mm	259	189
Potential evapotranspiration	mm	2542	2542

algorithm iteratively assigns elements to clusters until it converges to a stable solution, starting with k initial group centres. Observations are assigned to the group with the closest centre. The mean of the observations assigned to each of the groups is computed, and the procedure is repeated until every observation remains in the same group as in the previous iteration.

3.3 Conceptual model

The conceptual, semi-distributed rainfall–runoff model HBV light (Bergström 1976, 1995, Seibert 1997, Seibert and Vis 2012) was used in the study, as it provides a relatively simple approach to handling the spatial distribution of hydro-climatic variables, snowmelt processes and other catchment characteristics related to steep elevation gradients. HBV light uses four routines: a distributed snow routine, a soil moisture routine, a lumped response routine and a routing routine (Fig. 3). Snow accumulation and melt are calculated by the degree-day method at five elevation zones of roughly 500 m intervals. This temperature-based approach is practical, since the HBV light model is able to spatially distribute the climate variables without requiring comprehensive datasets for the upstream high altitudes (including climate, snow depth or snow water equivalent data), as required by physically-based energy balance modelling. Groundwater recharge and actual evapotranspiration are functions of water storage in the soil moisture routine. Finally, the lumped response function transforms recharge into discharge. A detailed description of the model can be found in Seibert and Vis (2012).

After testing different model structures in terms of number of parameters (from 8 to 18), we settled on the

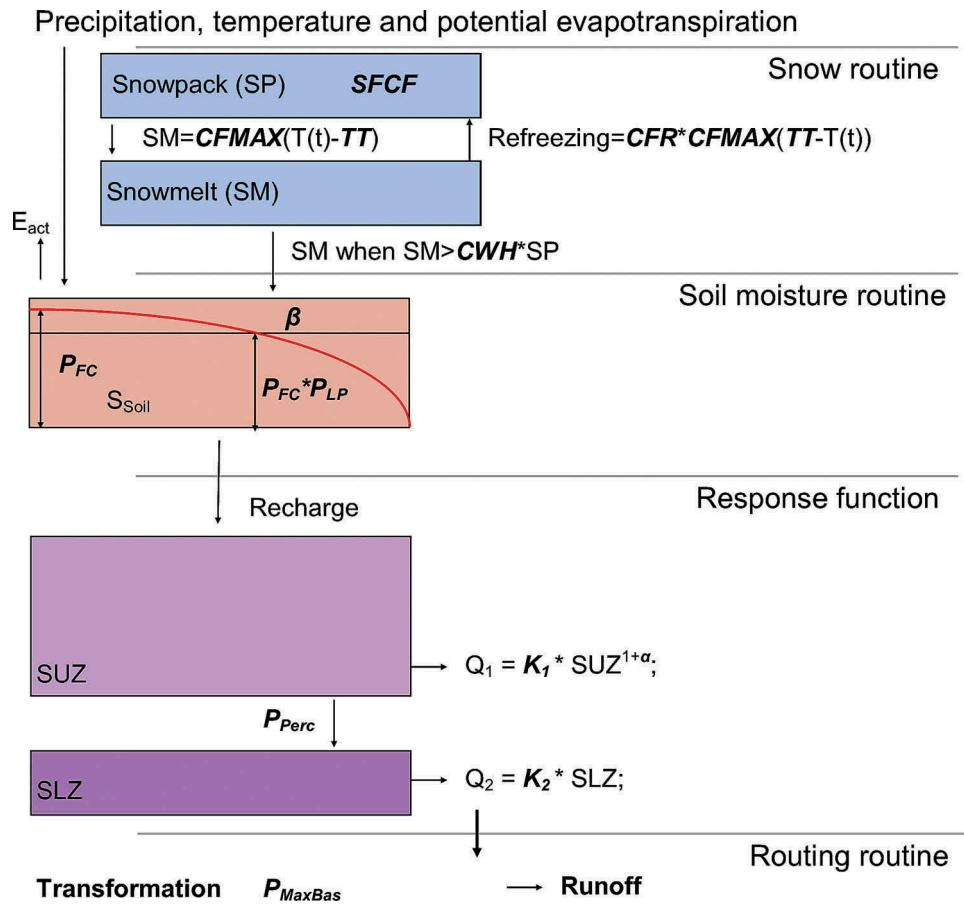


Figure 3. The structure of the HBV light model used with calibrated parameters (in red).

simplest model set-up, including two reservoirs in series and the snowmelt component. This structure involved 12 parameters, which are listed in Table 2, namely seven rainfall–runoff parameters, a calibrated precipitation gradient (one parameter) and four parameters for the snowmelt routine. The parameter LP, i.e. the threshold evaporation reduction, was fixed to 0.6 due to its insensitivity, while for the temperature altitudinal gradient we set a typical environmental lapse rate of $-0.6^{\circ}\text{C}/100 \text{ m}$.

In situ geochemical tracer and discharge data, provided by Nauditt *et al.* (2016), were consistent with the selection of the presented model structure, as they suggest that runoff sources in the catchments appear to be dominated by deeper flow paths from more slowly responding reservoirs. Initial model tests showed that two reservoirs were needed to adequately simulate the hydrograph, including a nonlinear upper storage component (parameters α , K_1), a recharging parameter (PERC), a lower linear storage component (parameter K_2) representing quick surface flow, and a slower groundwater runoff component.

The total runoff, which is sum of the two storage outflows Q_1 and Q_2 , is routed to the catchment outlet via a

triangular weighting function (parameter MAXBAS). The lumped soil moisture capacity (parameter FC) routine controls the conversion of ET_0 into actual ET (parameter LP), according to actual soil moisture availability (Seibert 2005) and the recharge to the runoff-generating storage (parameter β). The temperature (T_{calt}) and precipitation (P_{calt}) gradients were used to reflect the unknown and probably highly variable spatial distribution of rainfall across the elevation zones.

Due to the significant inter-annual climatic variability, the first three recorded years, 1968–1971, were used for the warm-up period prior to the study period, in order to provide an adequate spin up (1971–2013). An initial model calibration, using 30 years of data (i.e. 1971–2001) served as a baseline simulation to assess the overall model performance.

3.3.1 Model calibration

Random parameter sets were derived from one million uniform Monte Carlo (MC) sampling from *a priori* defined ranges. The model performance was evaluated using the Nash-Sutcliffe efficiency (NSE) for the goodness of fit of flow simulations, and its transformed expression considering the logarithm of discharge

Table 2. Parameters and associated model components.

Parameter	Explanation	Initial	Unit	Minimum	Maximum
<i>Snow routine</i>					
TT	Threshold temperature	(−3, 3)	°C	−3	3
CF _{max}	Degree-day factor	(1, 10)	mm °C ^{−1} d ^{−1}	1	10
SFCF	Snowfall correction factor	(0, 2)	–	0	2
CFR	Refreezing coefficient	(0.05)	–	0.05	0.05
CWH	Water holding capacity	(0, 0.5)	–	0	0.5
<i>Soil and evaporation routine</i>					
FC	Maximum SM (storage in the soil)	(1, 1000)	mm	1	1000
LP	Threshold for reduction of evaporation (SM/FC)	(0.6)	–	0.6	1
β	Shape coefficient	(1, 10)	–	1	10
PERC	Maximal flow from upper to lower GW-box	(1, 500)	mm d ^{−1}	1	500
α	Nonlinearity parameter	(0, 1)	–	–	–
<i>Groundwater and response routine</i>					
K ₁	Recession coefficient	(0.01, 0.5)	d ^{−1}	0.01	0.5
K ₂	Recession coefficient	(0.0001, 0.1)	d ^{−1}	0.0001	0.1
<i>Routing routine</i>					
MAXBAS	Routing, length of weighting function	(1, 10)	d	1	10
P _{calt}	Precipitation gradient	(1, 50)	%	1	50
T _{calt}	Temperature gradient	(0.6)	°C	0.6	1

(lnNSE; Seibert 2005), which is considered an appropriate objective function for low-flow modelling (Krause *et al.* 2005, Van Loon and Van Lanen 2012).

After comparing the MC-based calibration of the long-term baseline of 30 years (1971–2001) and a dry 10-year period (2003–2012), we used dry and wet sub-periods as selected by the cluster analysis for separate model calibrations. Each sub-period was preceded by a 3-year warm-up period.

Hydraulic turnover times were estimated based on the MC performance of parameter K_2 , representing the linear lower storage inferring slower groundwater runoff components. The parameter range was set from 0.0001 to 0.1 day^{−1}, which corresponds to a turnover time, T , ranging from 100 to 10 000 days, thus capturing the diversity of potential groundwater stores in mountain environments (e.g. Uhlenbrook *et al.* 1999). We remark that T is defined as the time taken to exchange the total mobile water volume, V_{mob} , of a hydrological system under steady-state conditions, i.e. $T = V_{\text{mob}}/Q$, where Q is the flow rate (Melhorn *et al.* 1998).

3.3.2 Parameter identifiability

The 1% best-performing parameter sets out of the one million MC sample, in terms of model efficiencies (NSE and lnNSE), were retained to represent the parameter uncertainty in the form of 5th/95th percentiles of the simulations. From the retained parameter sets, the cumulative distribution function was calculated, which gives an indication of parameter identifiability, similar to the method introduced by Hornberger *et al.* (1986). A curved or constrained parameter range indicates an accumulation of sensitive parameters, while a straight line represents an unidentifiable parameter.

4 Results

4.1 Identification of wet and dry periods

We clustered, through the k -means approach, the annual runoff data of the Rio Grande and the Tascadero headwater catchments for the hydrological years between 1969 and 2013 (Table 3). The results for both catchments were similar, indicating almost identical climatic drivers. The number of dry years was 21 for the Tascadero and 25 for the Rio Grande catchment. The dry cluster at the smaller Tascadero catchment ranges from 19.5 mm to 107.7 mm, with an average of 54.6 mm. The larger Rio Grande catchment showed a dry cluster with slightly higher values ranging from 55.9 to 174.2 mm (average 110.1 mm). For the Tascadero catchment, 9 years were clustered as wet, and for the Rio Grande, 8 years. Medium years numbered 14 for the Tascadero and 11 for the Rio Grande. The upper limit of the wet cluster was 580.0 mm at the Tascadero and 765.7 mm at the Rio Grande.

Figure 4 shows the distribution of wet, dry and medium years according to the k -means runoff cluster analysis for the Rio Grande catchment. The wettest

Table 3. K -means annual runoff cluster results for wet, medium and dry hydrological years for the Rio Grande (Las Ramadas station) and Tascadero catchments.

	Wet years	Medium years	Dry years
<i>Rio Grande</i>			
Lower limit:	449.2 mm	198.9 mm	55.9 mm
Average:	586.8 mm	277.5 mm	110.1 mm
Upper limit:	765.7 mm	388.3 mm	174.3 mm
	8 wet years	11 medium years	25 dry years
<i>Tascadero</i>			
Lower limit:	326.8 mm	129.2 mm	19.5 mm
Average:	449.2 mm	187.8 mm	54.6 mm
Upper limit:	570.5 mm	293.0 mm	107.7 mm
	9 wet years	14 medium years	21 dry years

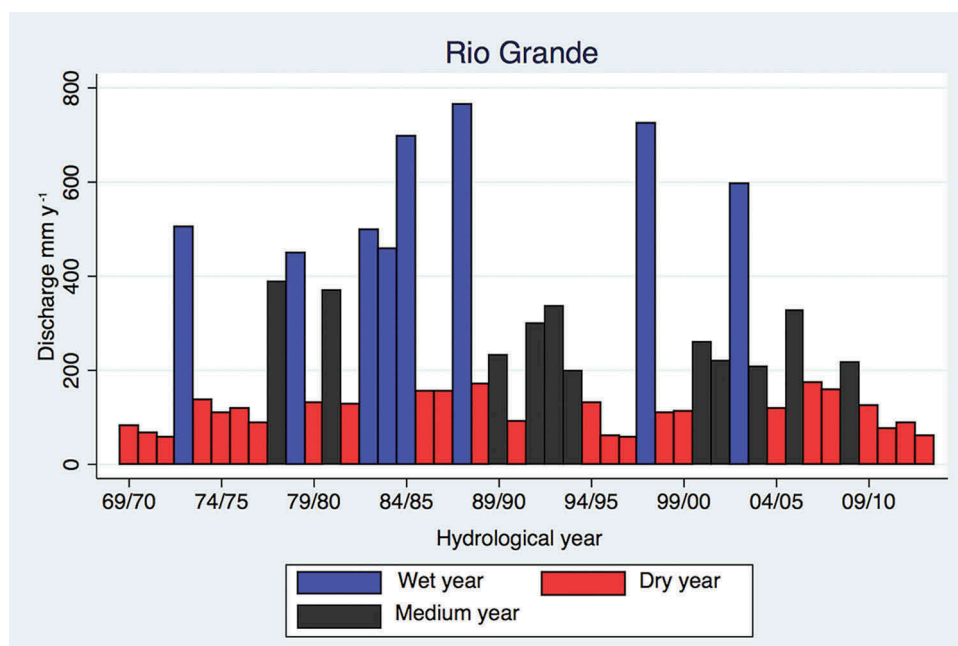


Figure 4. Discharge cluster results for individual hydrological years from 1968 to 2013 at Las Ramadas station (Rio Grande catchment).

hydrological years were identified as 1984 and 1997, which are related to El Niño years. However, it is apparent that wet years tend to be single extreme years, in contrast to dry years, which more frequently cluster with two or more consecutive years. Additionally, the frequency of wet years decreased after the 1980s, with only two wet years since 1986 and periods spanning decades (e.g. since 2000) without a single wet year (Fig. 4). On the other hand, a number of moderate years reached high annual runoff of up to 360 mm. Since the late 1980s, the number of moderate years has increased compared to the number of wet and dry years.

4.2 Sub-period model calibration and performance

The 30-year baseline calibration (1971–2001) showed reasonable overall results (Tables 4 and 5), which mostly matched the hydrograph (max NSE = 0.34, max lnNSE = 0.39) for the Rio Grande catchment, and medians for the retained parameter sets of NSE = 0.42 and lnNSE = 0.47 for the Tascadero catchment. Although lower than the efficiencies considered behavioural in conventional modelling studies, these values are considered acceptable given the uncertainties regarding the spatial and temporal variation in precipitation inputs and snowmelt outputs in an unmonitored montane catchment with such an extreme elevation gradient. Moreover, the single calibration period encompassed marked variations in the hydro-

climatic conditions. The calibrated parameters were evaluated against observed data for 2002–2013 and produced good results for the Rio Grande, with improved efficiency statistics, while the test for the Tascadero was less successful. However, a closer inspection revealed that the model was averaging through transitions from dry to wet periods and *vice versa*, with neither peak flows nor low flows being particularly well reproduced. This indicates that using constant model parameters is likely to either overestimate seasonal low-flow periods and/or underestimate high-flow periods. To test whether the use of different parameter sets for periods of contrasting hydrological behaviour improves the model performance, we used the sub-periods selected by the cluster analysis (Tables 3 and 4). These were then used as separate periods for model calibration, to better capture the impacts of hydro-meteorological variability on the hydrological functioning, as also reported by other researchers (e.g. Gharari *et al.* 2013, Westra *et al.* 2014, Thirel *et al.* 2015). This is illustrated using representative dry and wet sub-periods. As representative of dry conditions, three consecutive hydrological years between April 1973 and May 1977 were selected, while for wet conditions, the years between 1983 and 1986 were chosen. These two sub-periods were selected for model calibration based on the longest consecutive clusters of dry and wet years (Fig. 4).

Figure 5 shows the 5% and 95% uncertainty bounds compared to the observed discharge and precipitation

Table 4. HBV light MC ranges and calibration results for medium, wet and dry hydrological years for the Rio Grande catchment.

Parameter	Initial	Unit	Posterior median (5th, 95th)					Normal/wet 2000–2003
			1971–2001 Long-term period	Dry 2003–2013 Dry period	Dry 1973–1977	Wet 1983–1986	Wet/normal 1991–1994	
TT	(–3, 3)	°C	0.4 (–1.4, 1.5)	–0.76 (–2.8, 2.6)	–1 (–2.6, 2.8)	–0.24 (–2.7, 1.8)	1.6 (–0.23, 2.35)	–0.75 (–2.7, 1.3)
CF _{max}	(1, 10)		7.1 (3.6, 9)	6.3 (1.7, 9.3)	6.8 (2, 9.6)	7.9 (4.7, 9.9)	8.3 (5.2, 9.6)	7.7 (3.2, 9.6)
SFCF	(0, 2)		0.83 (0.34, 1.07)	0.2 (0.05, 0.52)	0.61 (0.07, 1.85)	0.88 (0.49, 1.95)	1.2 (0.6, 1.8)	1.4 (0.4, 1.85)
CFR	(0.05)							
CWH	(0, 0.5)		0.26 (0.07, 0.46)	0.24 (0.02, 0.48)	0.28 (0.02, 0.45)	0.21 (0.02, 0.46)	0.3 (0.04, 0.49)	0.29 (0.1, 0.46)
FC	(1, 1000)	mm	23.9 (21, 44.2)	113 (14.3, 428)	460 (53, 921)	10.4 (1.6, 38.2)	19.6 (8.6, 58)	32.3 (4.7, 215)
LP	(0.6)	–						
β	(1, 10)	–	2.35 (1.2, 7)	3.4 (1, 9.1)	4.7 (1.4, 9.5)	3.9 (1.2, 8.8)	3.4 (1, 9)	2.3 (1, 8.6)
PERC	(1, 500)	mm d ^{–1}	86.2 (49.8, 97.6)	338 (119, 458)	206 (80, 468)	249 (45, 496)	263 (121, 444)	208 (31, 446)
α	(0, 1)	–	0.27 (0.03, 0.95)	0.44 (0.04, 0.97)	0.52 (0.06, 0.92)	0.61 (0.04, 0.94)	0.45 (0.12, 0.82)	0.61 (0.1, 0.94)
K_1	(0.01, 0.5)	d ^{–1}	0.1 (0.015, 0.34)	0.23 (0.03, 0.48)	0.17 (0.02, 0.49)	0.27 (0.03, 0.46)	0.3 (0.07, 0.49)	0.29 (0.06, 0.46)
K_2	(0.0001, 0.1)	d ^{–1}	0.0086 (0.008, 0.009)	0.0003 (0.0001, 0.0005)	0.0004 (0.0003, 0.0007)	0.01 (0.008, 0.15)	0.008 (0.005, 0.012)	0.01 (0.01, 0.018)
MAXBAS	(1, 10)	d	6.1 (2.7, 9.1)	5.2 (1.8, 9.4)	5.5 (2.2, 9.5)	5.2 (1.4, 9.3)	5.1 (1.2, 9.6)	5.4 (1.8, 9.1)
P_{calt}	(1, 50)	%	2.7 (1.2, 10.7)	19.8 (3.1, 44.8)	16.9 (4.1, 38.8)	10.7 (2.4, 24.9)	9.1 (2.2, 26.8)	15.8 (3.3, 43.3)
T_{calt}	(0.6)	°C						
Performance								
NSE		–	0.19 (0.08, 0.34)	0.31 (0.2, 0.43)	0.11 (0.11, 0.14)	0.68 (0.61, 0.75)	0.57 (0.41, 0.68)	0.67 (0.61, 0.79)
InNSE		–	0.22 (0.11, 0.39)	0.39 (0.31, 0.53)	0.14 (0.11, 0.17)	0.65 (0.6, 0.74)	0.46 (0.4, 0.6)	0.68 (0.61, 0.74)

Table 5. HBV light MC ranges and calibration results for medium, wet and dry hydrological years for the Tascadero catchment.

Parameter	Initial	Unit	Posterior median (5th, 95th)					Normal/wet 2000–2003
			1971–2001 Long-term period	Dry 2003–2012 Dry period	Dry 1973–1977	Wet 1983–1986	Wet/normal 1991–1994	
TT	(–3, 3)	°C	1 (–1.9, 2.5)	–0.44 (–2.1, 2.3)	1.74 (–2.9, 2.6)	0.15 (–1.9, 1.6)	1.1 (–1.9, 2.38)	0.59 (–1.9, 2.3)
CF _{max}	(1, 10)		7.9 (3.7, 9.3)	4.9 (1.6, 9.7)	7.1 (2.9, 9.5)	7 (3.6, 9.7)	5 (2, 9.7)	5.2 (1.9, 8.6)
SFCF	(0, 2)		0.97 (0.72, 1.39)	0.08 (0.02, 0.18)	0.42 (0.25, 0.71)	0.33 (0.19, 0.7)	0.97 (0.48, 1.7)	0.32 (0.18, 0.79)
CFR	(0.05)							
CWH	(0, 0.5)		0.28 (0.03, 0.46)	0.26 (0.02, 0.46)	0.25 (0.08, 0.48)	0.25 (0.07, 0.44)	0.26 (0.04, 0.46)	0.24 (0.04, 0.45)
FC	(1, 1000)	mm	226 (49, 837)	184 (3.8, 945)	273 (21, 907)	105 (30, 747)	576 (83, 894)	567 (129, 991)
LP	(0.6)	–						
β	(1, 10)	–	1.4 (1.1, 2.9)	2.9 (1.1, 7.9)	1.2 (1, 7.3)	1.9 (1.1, 7.1)	1.5 (1.1, 3.4)	1.24 (1, 1.7)
PERC	(1, 500)	mm d ^{–1}	281 (98, 483)	262 (72, 471)	269 (58, 496)	215 (20, 483)	272 (95, 467)	352 (148, 486)
α	(0, 1)	–	0.28 (0.12, 0.88)	0.5 (0.12, 0.91)	0.5 (0.08, 0.94)	0.45 (0.03, 0.97)	0.56 (0.08, 0.94)	0.58 (0.03, 0.98)
K_1	(0.01, 0.5)	d ^{–1}	0.25 (0.08, 0.47)	0.36 (0.06, 0.47)	0.35 (0.07, 0.48)	0.22 (0.1, 0.48)	0.22 (0.03, 0.47)	0.32 (0.16, 0.49)
K_2	(0.0001, 0.1)	d ^{–1}	0.008 (0.007, 0.01)	0.003 (0.0006, 0.006)	0.006 (0.005, 0.009)	0.01 (0.009, 0.015)	0.01 (0.008, 0.015)	0.01 (0.008, 0.02)
MAXBAS	(1, 10)	d	4.9 (1.8, 9)	6.9 (1.3, 9.6)	6 (1.3, 9.8)	6.2 (1.6, 9.5)	5 (1.7, 9.3)	3.2 (1.8, 8.7)
P_{calt}	(1, 50)	%	48 (1.1, 9.4)	23.8 (5.1, 45.6)	10 (2.1, 22.6)	24 (10, 38)	5.5 (1.7, 16.5)	18.6 (4.6, 39)
T_{calt}	(0.6)	°C						
Performance								
NSE		–	0.42 (0.4, 0.44)	0.09 (0.04, 0.23)	0.1 (0.06, 0.17)	0.53 (0.5, 0.56)	0.52 (0.5, 0.58)	0.64 (0.61, 0.76)
InNSE		–	0.47 (0.43, 0.56)	0.13 (0.11, 0.19)	0.26 (0.21, 0.34)	0.52 (0.5, 0.55)	0.41 (0.32, 0.59)	0.43 (0.41, 0.49)

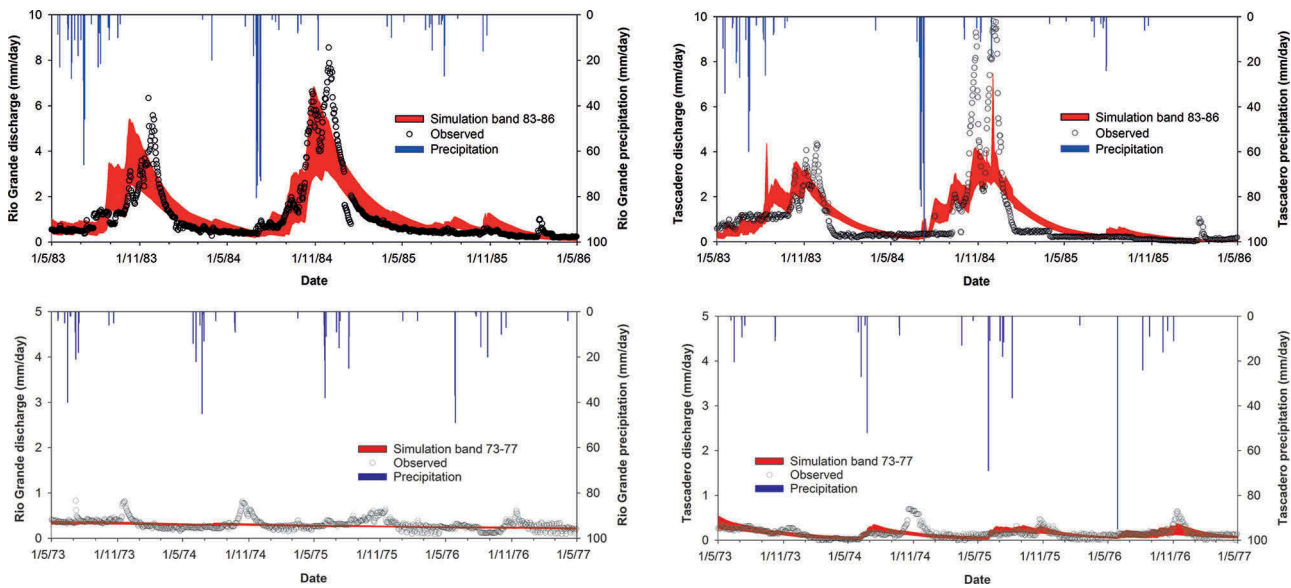


Figure 5. Discharge simulations for the wet (1983–1986) and the dry period (1973–1977) at the Rio Grande and Tascadero catchments.

for the dry and wet periods in both catchments. The wet period calibration showed a median best fit of $NSE = 0.68$ and $\ln NSE = 0.65$ for the Rio Grande, and median $NSE = 0.53$ and $\ln NSE = 0.52$ for the Tascadero. However, the dry period calibration showed best performance values of only $NSE = 0.11$ and $\ln NSE = 0.14$ for the Rio Grande, and $NSE = 0.10$ and $\ln NSE = 0.26$ for the Tascadero.

For wet periods, the model captured the general dynamics, by reflecting the baseflow-dominated low-flow season and the snowmelt-dominated high-flow season. However, the peak flows were still not well matched and a timing error showed an early snowmelt onset in the model. As shown in Figure 5(a), there are also evident data errors, such as abrupt changes (e.g. in May 1985,) that were not corrected prior to modelling due to unknown error sources. For the wet period, the peak flows were underestimated compared to the observed ones. Especially for the Tascadero catchment, the simulated daily peaks from October to January only reached 4 mm, while the observed peaks were as high as 10 mm. The differences for the Rio Grande are smaller, since the measured daily peak runoff was 8.2 mm and the simulated one was 6.3 mm. Furthermore, in both catchments the simulated discharge recession at the end of the melting season from December to January was much slower than the more abrupt observed decrease, which was particularly marked in the Tascadero. This may reflect less extensive aquifer systems in the smaller, as well as structurally simpler, Tascadero catchment. The baseflow was simulated quite well for the wet period in both catchments, with

only slightly overestimated values at the end of the melting season.

In both catchments, the model was able to simulate the baseflow for the entire dry period (1973–1977); however, the small peaks of up to a maximum of 0.8 mm during the melting season were not well captured (Fig. 5). The very low precipitation rates generated discharges of below 0.1 mm with almost no flow dynamics, although the modelled volumes were reasonable. Such poor model performance to a large extent reflects the lack of signal in the flow data during dry years.

4.3 Parameter identifiability for wet and dry periods

The parameter sensitivity was assessed to compare identifiability across the wet and the dry seasons. The 1% best-performing parameter sets out of 10^6 MC sets were retained for this analysis. Most of the rainfall-runoff parameters could be constrained during model calibration, but did not show significant differences between the selected sub-periods and were similar for the two catchments (Tables 4 and 5). For example, for the nonlinearity parameter α we obtained median values of 0.52 for the dry and 0.61 for the wet period at the Tascadero; the corresponding medians for the Rio Grande were 0.50 and 0.45. However, for the 30-year calibration period, parameter α showed lower nonlinearity, with a median value of 0.27, which reflects the averaging of variability mentioned above.

The medians for the upper storage parameter K_1 were 0.17 and 0.27 for the Rio Grande and 0.35 and 0.22 for the Tascadero, for the dry and wet periods, respectively. The calibrated precipitation gradient P_{calt} had higher values for the dry period in the Rio Grande (median of 16.9%, minimum 4.1%, maximum 38.8%) and lower values (10.7%, 2.4%, 24.9%) for the wet calibration sub-periods. For the 30-year period, P_{calt} resulted in much lower calibrated percentages with a median of 2.7% (1.2–10.7%) for the Rio Grande and a median of 4.8% (1.1–9.4%) for the Tascadero, which suggests a more averaged-out long-term simulation, thus justifying the introduction of sub-periods. Also, most of the snowmelt routine parameters did not indicate sensitivity across the wet and dry periods.

However, the sub-period calibration procedure did reveal three parameters with markedly different values for the wet and the dry periods (strikingly, these have different median values and the 95% percentile ranges do not overlap). For the Rio Grande, these are the snowfall correction factor, SFCF, the maximum soil moisture storage, FC, and the groundwater storage coefficient, K_2 (Table 6). The identifiability of these key parameters is visualized in Figure 6. This shows that these parameters are different for the dry and wet periods due to the contrasting characteristics of their CDFs in terms of no overlap, different curvature and constrained parameter ranges. The K_2 showed lower values almost systematically for the dry period (median of 0.0004) compared to the wet period (median of 0.01). This would be consistent with a greater recharge rate during wet conditions and a more marked groundwater flux to the stream (Fig. 6(a)). The FC in the calibration range between 1 and 1000 mm varied from a median of 10.4 mm during the wet period up to a median of 460 mm during the dry period. Hence, the recharge to the upper reservoir was conceptualized as being much lower during the dry season, which is consistent with the reduced recharge to

the lower groundwater under dry conditions (Fig. 6(b)). Finally, the SFCF compensates for systematic errors in the snowfall estimates, with a higher median value for the wet season (0.88) and a lower median value (0.61) for the dry season.

The analysis for the Tascadero catchment showed lower sensitivity of the associated parameters, since only K_2 could be distinguished for wet and dry periods. Similarly to the Rio Grande, during the wet season K_2 was much higher (0.01) compared to the dry season value (0.006; Table 4). However, the dry season values of K_2 were higher for the Tascadero than for the Rio Grande (median of 0.0004). The minor differences between the two catchments could be related to their differences in terms of size and elevation, as well as unaccounted differences in their hydro-geological regimes.

4.4 Modelled storage dynamics and turnover times

The model structure utilized three storage components. However, simulations showed that the water fluxes were almost exclusively generated from the lower groundwater reservoir, recharged by a seasonally intermittent soil moisture store of up to 30 mm during the melting season (September to November). The groundwater reservoir has storage of several hundred millimetres to sustain flows during both wetter and drier periods. For instance, during the wet hydrological year 2002/03, up to 430 mm of groundwater was recharged and stored, compared to less than 50 mm during dry years (2006/07). Unfortunately, with our sub-period calibration approach we cannot allow the accumulation of storage deficits over successive years, though in most cases these were replenished by recharge during the melt season.

The storage dynamics are directly linked to the hydraulic turnover times, which were different for dry and wet periods (Table 7). While for the Rio Grande, during the dry period from 1973 to 1977 a median turnover time of 2500 days, with an upper boundary of 3333 days and a lower boundary of 1429 days, was identified, for the wet period (1983–1986) a median of only 100 days was estimated. This faster turnover of storage during wetter periods, with reduced residence times, has been commonly observed in upland catchments (Birkel *et al.* 2015).

Although the wet season turnover times were similar for both catchments, the dry season values for the smaller Tascadero catchment were much lower (median 333 days), compared to the median of 3333 days for the larger Rio Grande catchment (Table 6). Although it should be noted that some retained

Table 6. Sub-period sensitive parameters for the Rio Grande and Tascadero catchments.

Parameter	Initial	Unit	Posterior median (5th, 95th)	
<i>Rio Grande</i>				
SFCF	(0, 2)	–	Dry 1973–1977 0.61 (0.07, 1.85)	Wet 1983–1986 0.88 (0.49, 1.95)
FC	(1, 1000)	mm	460 (53, 921)	10.4 (1.6, 38.2)
K_2	(0.0001, 0.1)	d ^{−1}	0.0004 (0.0003, 0.0007)	0.01 (0.008, 0.15)
<i>Tascadero</i>				
K_2	(0.0001, 0.1)	d ^{−1}	Dry 1973–1977 0.006 (0.005, 0.009)	Wet 1983–1986 0.01 (0.009, 0.15)

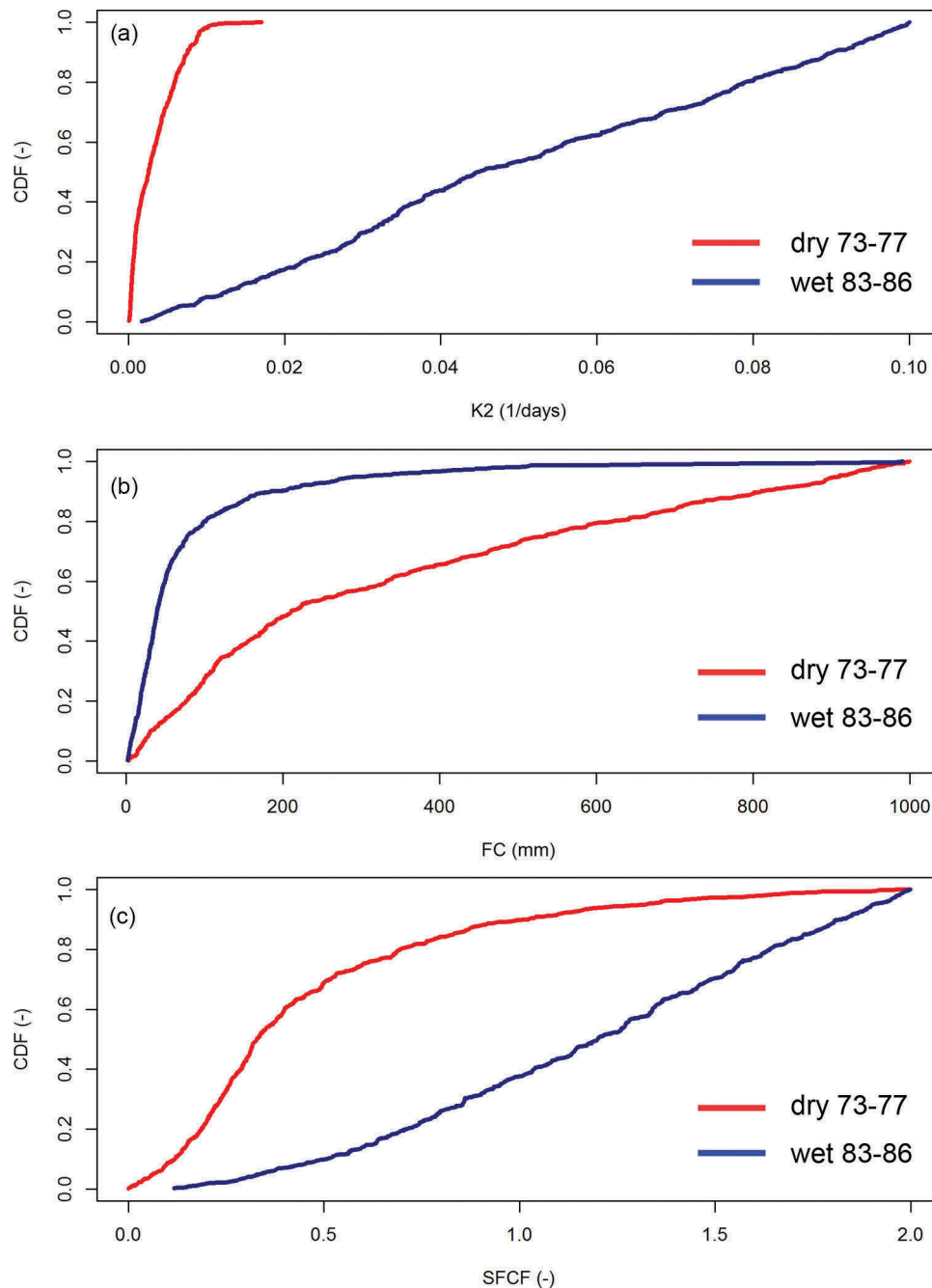


Figure 6. Comparison of parameter identifiability for the wet and the dry periods for the Rio Grande for parameters (a) K_2 , (b) FC and (c) SFCF.

Table 7. Hydraulic turnover times for the Rio Grande and Tascadero catchments across different periods.

Catchment	Unit	Hydraulic turnover posterior median (5th, 95th)					
		1971–2001	Dry 2003–2013	Dry 1973–1977	Wet 1983–1986	Wet/normal 1991–1994	Normal/wet 2000–2003
Rio Grande	d	116 (111, 125)	3333 (2000, 10000)	2500 (1429, 3333)	100 (7, 125)	100 (50, 200)	100 (56, 100)
Tascadero	d	125 (100, 143)	333 (167, 1667)	167 (111, 200)	100 (7, 111)	100 (70, 125)	100 (80, 125)

parameter sets gave dry period turnover times of longer than 1000 days, this difference may be related to hydro-geological characteristics, with a more complex geology with higher storage in the Rio Grande,

which may be evident in the more complex catchment structure. This would also help explain the much steeper recession in the Tascadero, as mentioned above (Fig. 5).

5 Discussion

5.1 How well can a conceptual, semi-distributed model capture the influence of climatic gradients and hydro-climatic variability on runoff generation in the semi-arid Andes?

A semi-distributed configuration of the conceptual HBV light model was applied to capture the elevation gradient in precipitation in two perennial Andean headwater catchments. After testing alternative model structures applied to the baseline calibration period from 1971 to 2001, a simple model based on a snowmelt routine and two runoff generation reservoirs was adopted. The model structure and parameterization were complemented based on seasonal synoptic empirical field observations, including hydro-chemistry and stable isotopes, meteorological and discharge measurements. The challenging range of hydro-meteorological conditions over a 30-year record resulted in a single calibration of the model averaging out the marked variability and failing to capture both the highest and lowest flows adequately, limiting such an approach in terms of aiding water management. Therefore, representative dry and wet sub-periods were selected for separate calibrations, based on statistical *k*-means runoff clustering of hydrological years 1969–2013. The dry period from 1973 to 1977 and the wet period from 1983 to 1986 were modelled with 10^6 MC parameter sets, from which the 1% best-performing were retained to show the resulting 5th/95th percentiles. Although not a formal uncertainty assessment, the simulation bounds give some indication of the uncertainty propagated through the model due to uncertain data inputs (unknown precipitation gradient and snow inputs), model structural errors and parameter uncertainty. The data used for calibration have some limitations in terms of quality, and are highly likely to include errors that make them unsuitable for calibration (Beven and Westerberg 2011). However, these problems are typical for most model applications, especially in sparsely gauged, remote basins.

As is usual in rainfall–runoff modelling, the best simulations from the retained parameter sets for the wet period performed much better than those for the dry period (Pilgrim *et al.* 1988, Lidén and Harlin 2000). The model efficiencies for the wet season, with a median NSE value of 0.68 and an upper limit of 0.75, showed good results for semi-arid catchments, where baseflow needs to be simulated for dry periods, despite the very few rainfall events in the region and difficulties in representing the groundwater storage over long periods (see Section 4.5; see also van Loon and van Lanen

2012). Although the water balance is forced to be closed on an annual basis by the HBV model, there is a trade-off between calibrating the baseflow-generating parameters controlling the groundwater recharge and turnover and simulation of the snowmelt-driven peak.

5.2 Is sub-period calibration useful to assess dry and wet periods?

Sensitive parameters have been identified through examining the differences in parameter values within the 1% best-performing parameter sets. For the Rio Grande catchment, a significant difference for dry and wet periods was detected in the snowfall correction factor, SFCF, the maximum soil moisture storage, FC, and the groundwater storage coefficient, K_2 (Table 6). Such variability clearly shows that the optimized model parameters are not constant between contrasting periods. Westra *et al.* (2014) and Tetzlaff *et al.* (2014) used a similar approach to examine non-stationarity in their modelling. Sub-period calibration was also used by Gharari *et al.* (2013) to further select time-consistent model parameter sets, albeit in a more humid and climatically more homogeneous catchment. In addition to this variability in model parameter sets, a difference between the sub-periods reflects inherent data uncertainties. For example, the variation in the SFCF value reflects the uncertainty regarding precipitation input to the catchment, which is expected to vary spatially in high mountains. Furthermore, wind speeds are extremely high in the Cordillera (Kalthoff *et al.* 2002), with potential related impacts on the redistribution of snow and the water balance; these are not considered in the model. In addition to the unknown high-altitude snow input, the varying but insensitive precipitation gradient, P_{calt} , reflects the necessary yet unknown climate gradient (median values vary between 16.9% and 10.7%). Similarly, the time-varying runoff parameters, such as the soil storage FC and K_2 , have differences of almost an order of magnitude between dry and wet periods, a range of variability that cannot be covered with traditional time-invariant calibration strategies. The latter is also why the model applied to the whole period of record performs much worse than the sub-period simulations (Tables 4 and 5). A way forward could be to introduce dynamic (i.e. time-varying) model parameters based on a few key parameters that are well identified (particularly K_2), and to constrain the range of variability of these using the sub-period calibration results (Efstratiadis *et al.* 2015).

5.3 What information could be gained to inform water resources management in the area?

A key objective of this study was to develop a model that could give a reasonable prediction of flows and hence water resources availability across periods of remarkable hydro-climatic variability. Such short-term predictions would allow streamflow forecasts to be used for the melting season (October–December) and the dry season (September–May), based on precipitation during the wet season (May–August). Long-term predictions would also allow assessment of the impacts of expected climatic variability on inter-annual variability of water availability. According to IPCC (2013), in the long term (i.e. 2081–2100), the mean annual temperatures in central Chile are expected to increase by 1–4°C, with daily increases of up to 5°C. In addition, the precipitation rates are expected to decrease by around 20% between March and May. Developing long-term discharge predictions based on such projections is challenging, due to the range and uncertainties related to climate modelling and regional downscaling, especially in mountainous regions (Randall *et al.* 2007).

Alternatively, seasonal discharge predictions based on winter precipitation and exploratory long-term scenarios could be based on estimated input data from historical records, which reflect consistently higher mean daily temperatures and precipitation extremes from the wet and dry periods investigated, corresponding to climate change projected temperature increases and expected extremes. Hence, the dry and wet sub-period calibration clearly demonstrated the importance of using distinct model parameter sets associated with periods of hydro-climatic contrasts (Westra *et al.* 2014). The latter could be directly accounted for to create scenarios that are useful to water managers in central Chile, who currently have an urgent need for robust tools to assess the availability of water resources.

6 Conclusions and outlook

Dry conditions are historically dominant in northern central Chile and are expected to become more frequent in the future (Souvignat *et al.* 2010, Vicuña *et al.* 2010, IPCC 2013). Seasonal and long-term streamflow predictions require reliable models and further observations to inform water managers. For the first time, a conceptual model, namely HBV light, was applied on a daily basis to this high-altitude semi-arid environment, specifically in two gauged catchments in the Andean Cordillera. The model was initially calibrated for a 30-year period, and then, based on results from Monte Carlo simulation, it was re-calibrated across sub-

periods with consecutive wet and dry conditions. We found that the best simulations could be achieved with separate parameter sets for varying hydro-climatic conditions. The resulting snowmelt and groundwater model is a plausible representation of the hydrological cycle in these complex, data scarce Andean catchment systems. Lumped to a monthly scale, the model may provide useful discharge projections to water managers. However, it is clear that such a conceptualization is simplistic and there is an urgent need for more process-based approaches to understand the hydrology of this complex environment in a more comprehensive way.

In particular, high-altitude climate and snow input data from the Andean Cordillera are needed to improve the model and further understanding of mountain hydrology. The latter also has the potential to reduce model uncertainty. Additional tracer data of high temporal and spatial resolution would further help to improve process understanding, thus recognizing how these catchments transport and store water. In this context, it is also essential to improve representation of inter-annual accumulations of storage excesses and deficits, which are of key importance in the case of highly varying hydro-climatic regimes.

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